Humanity is poised for its next great expansion.

But our evolution has been confined to a single planet. To thrive beyond Earth, we must first navigate the final frontier: our own biology.

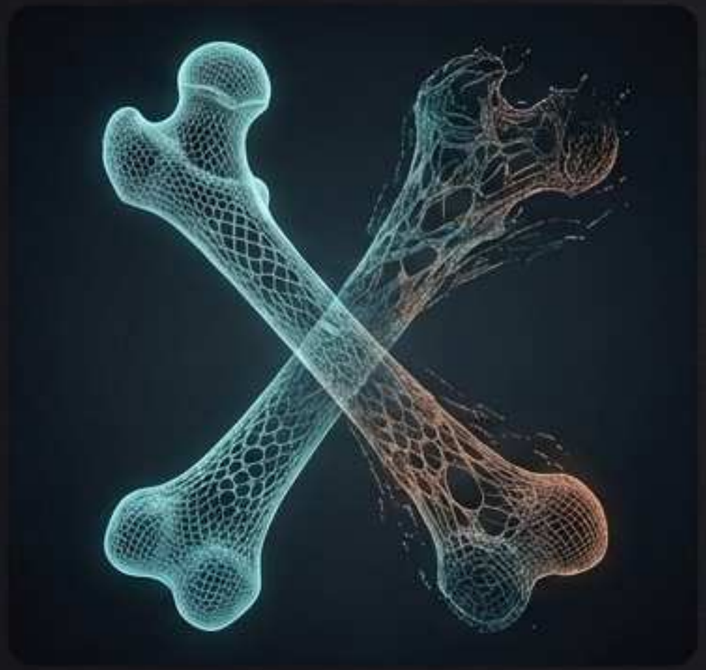
NotebookLM

The Unforgiving Environment of Space

Cosmic Radiation

- Exposure: 50-2000 mSv/year
- Consequences: DNA damage, cancer risk, cellular senescence

Microgravity



- Consequences: 1-2% monthly bone density loss, muscle atrophy (up to 20% in 6 months), cardiovascular deconditioning

Neuro-Cognitive Stress

- Sources: Isolation, confinement, circadian disruption
- Consequences: Impaired performance, depression

NotebookLM

The Blueprint for Resilience is Already in Our Genome

Human populations contain naturally occurring genetic variants that confer protection against extreme conditions. By systematically identifying these variants, we can map the pathways to enhanced resilience for spacefarers.

Our Approach: Move from population-level discovery to actionable therapeutic targets.

NotebookLM

Introducing the Spacefarer Phenome.

An integrated genomics platform to identify genetic variants and druggable targets that enhance human resilience in space.

We are building the genomic compass for humanity's journey to the stars.

NotebookLM

A Four-Stage Pipeline from Discovery to Action.

1. GWAS-Based Gene Discovery

Querying curated databases (UK Biobank, GWAS Catalog) for statistically significant associations ($P < 5\times10^{-8}$).

2. Mendelian Randomization

Using genetic variants as instrumental variables to distinguish causal relationships from correlation.

3. Phenome-Wide Analysis (PheWAS)

Identifying pleiotropic effects and potential off-target consequences across 500+ phenotypes.

4. Druggability Assessment

A multi-dimensional scoring system to identify targets with therapeutic potential.

NotebookLM

From Near-Term Application to Long-Term Transformation

Near-Term

Long-Term

Near-Term Applications

- Astronaut Selection
- Personalized Risk Assessment
- Hypothesis Generation

Long-Term Vision

- Preventive Interventions
- Gene Therapy
- Colonization Strategies

NotebookLM

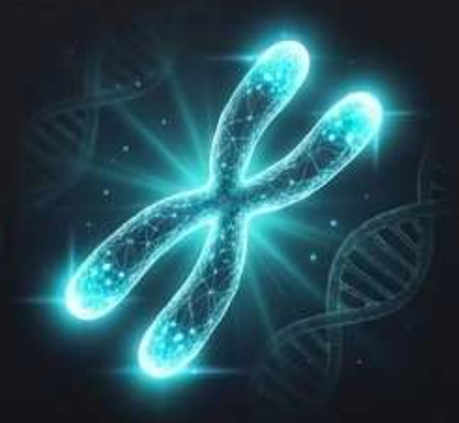
The Methodology Generalizes Beyond Space

The platform for optimizing human genetics for extreme environments has broad applications for challenges here on Earth

Deep-Sea Exploration

Arctic/Antarctic Habitation

High-Altitude Physiology

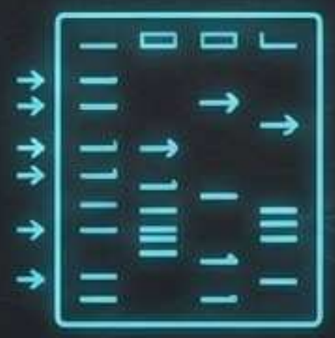


Longevity and Healthspan Extension

NotebookLM

We Are Seeking Partners to Build This Future.

We welcome collaboration and expertise in:



Genomics & Biostatistics

Space Medicine & Clinical Interpretation

Structural Biology & Druggability

Computational Biology & Pipeline Optimization

Ethics & Policy

NotebookLM

Building the genomic foundation for humanity’s expansion beyond Earth.

github.com/raimbetov/spacefarer-phenome

NotebookLM